

Concurrent and Distributed Systems

Lab 2 – Synchronisation in Java

Tasks

- Read the slides **Synchronisation in Java (Basics).pptx** and try the example programs shown in the slides (you can find them in **cds.lab2.zip**). Extend **Buffer** and **BoundedBuffer** by adding a new main class/method and threads that use these buffers.
- Learn by working through the examples in Java 9 Concurrency Cookbook: **Chapter 2. Basic Thread Synchronization** (see below).
 - Each example is introduced with a step-by-step explanation of how to write the Java classes, using the provided code fragments.
 - There are also GitHub repositories and the publisher's website that provide the complete Java classes for each book chapter example. However, the best way to learn is by building these programs yourself while reading the explanations in the book.
 - For each example, the book gives you the key ingredients, along with useful **Notes** and **Tips** to help you learn more about the Java concurrency framework.
 - If you get stuck on any example, please ask for help.

Java Concurrency Cookbook - Chapter 2. Basic Thread Synchronization

During the lab session it might be difficult to cover all the examples from Chapter 2. You can start with those highlighted in **bold and with [*]**, and continue with the remaining examples later.

- **Synchronizing a method [*]**
- **Using conditions in synchronized code [*]**
- **Synchronizing a block of code with a Lock [*]**
- **Synchronizing data access with read/write locks [*]**
- Using multiple conditions in a Lock
- Advanced locking with the StampedLock class

Solutions

The solutions are provided in **cds.lab2.zip**. The archive contains:

- The examples from the lecture slides
- All recipes from Java 9 Concurrency Cookbook - Chapter 2 (with minimal refactoring of the original source code).

Resources

- J. Fernandez Gonzalez (2017), **Java 9 Concurrency Cookbook** (2nd edition), Packt Publishing. You can access the online version through the library (log in with your university account), link [here](#). The GitHub repository providing the complete Java classes for each book chapter example is here: <https://github.com/PacktPublishing/Java-9-Concurrency-Cookbook-Second-Edition>

- J. Friesen (2015), **Java threads and concurrency utilities**, Apress. You can access the online version through the library (log in with your university account), link [here](#).
- B. Wittman, T. Korb, A. Mathur (2013), **Start Concurrent: An Introduction to Problem Solving in Java with a Focus on Concurrency**, Purdue University Press. Book also available in HTML format here <https://start-concurrent.github.io/>. You can jump directly to [Chapter 13. Concurrent Programming](#) or to any other Java topics you want to revise.
- A good video explaining multithreading, mutual exclusion and locks:
https://www.youtube.com/watch?v=7ENFeb-J75k&ab_channel=Computerphile